

ML Specification Report — playground-series-s6e2

Heart Disease Prediction · our from-scratch agent (GBDT pool + tree-search v3)

Figures grounded in the competition's `facts.json`, `config.yaml`, `eda_summary.json`, `STATUS.md`, and `experiments_tree_v3.json`.

Overview

The task is to rank 270,000 patients by heart-disease probability (metric: ROC-AUC) on a synthetic dataset derived from UCI Heart Disease. We solve it with a from-scratch pipeline — a pool of gradient-boosted trees (LightGBM / XGBoost / CatBoost) refined by a four-stage ladder and a 60-node tree search that ends on a 37-member mega-blend. Our champion scores **CV OOF AUC 0.955529** (submission file rounds to 0.95553); the NVIDIA reproduce-agent, copying a public XGBoost/LightGBM/CatBoost k-fold kernel, reaches **0.9554** — effectively a **tie**, with our agent nudging fractionally ahead.

Why it matters. Cardiovascular disease is a leading cause of death worldwide; risk-ranking from routine clinical features supports triage and preventive screening — as decision support, not diagnosis.

Data

Purpose of Data. Predict whether a patient has heart disease (Presence vs Absence) — a binary-classification Playground Series episode (S6E2). **Data Format** is clean **tabular CSV**, all numeric columns. **Data Volume** is **630,000 training rows / 270,000 test rows**, with **13 raw predictive features** once `id` and the target are removed.

Data Quality is high: **no missing values** in either split, **0 duplicate rows**, and only modest train ↔ test mean shift (largest $\approx +0.83\%$ on `Number of vessels fluoro`). The class balance is mild — **44.834% positive** (`Heart Disease` = Presence), an **imbalance ratio of 1.23** — which keeps the metric (AUC) clean and makes class-weighting largely unnecessary (confirmed in training). The strongest EDA signals, all consistent with cardiology intuition, were:

- **Thallium** is the dominant raw predictor (point-biserial $r \approx +0.606$).
- **Chest pain type** ($r \approx +0.461$), **Exercise angina** ($r \approx +0.442$), **Max HR** ($r \approx -0.441$), **Number of vessels fluoro** ($r \approx +0.439$), and **ST depression** ($r \approx +0.431$) form the strong mid-tier.
- **Slope of ST** ($r \approx +0.415$) and **Sex** ($r \approx +0.342$) contribute moderately.
- **BP** is essentially uninformative ($r \approx -0.005$); **FBS over 120** ($r \approx +0.034$) and **Cholesterol** ($r \approx +0.083$) are weak — a notable contrast with textbook risk-factor expectations.

Annotation Guidelines. The label is `Heart Disease` $\in \{\text{Absence, Presence}\}$, mapped to $\{0, 1\}$ (1 = Presence). Submissions are **positive-class probabilities** scored by AUC — a ranking, not a hard 0/1 decision.

Feature Set. Feb's focused **33-feature** set (13 raw UCI columns + 20 engineered) is carried into the July run and used for the XGBoost/CatBoost pool members; the 20 engineered columns were, however, **rejected for the LGB member** by a same-folds prior check (see the Training section), so the champion LGB family runs on the **13 raw features** alone:

Group	Features
Raw UCI (13)	Age, Sex, Chest pain type, BP, Cholesterol, FBS over 120, EKG results, Max HR, Exercise angina, ST depression, Slope of ST, Number of vessels fluro, Thallium
Thresholds / bins	Age_decade, BP_high, Chol_high, Thallium_7
Heart-rate derived	HR_reserve, HR_pct_max
Interactions	Age_x_MaxHR, Age_x_STdep, Age_x_Vessels, Chol_x_Age, BP_x_Chol, BP_x_Age, STdep_x_Slope, STdep_x_MaxHR, Vessels_x_Thallium, Angina_x_STdep, ChestPain_x_Angina
Composite / ratios	Risk_score, Chol_per_Age, BP_per_Age

Splitting strategy. A single canonical **Stratified 5-Fold CV (seed = 42, shuffle)**, shared across all four stages so scores are directly comparable. Because `StratifiedKFold` 's split depends only on the target vector and seed, the July folds reproduce the Feb v1 folds **byte-identically**, letting the reused Feb anchor sit on the same ladder. The leaderboard is closed (late submission returns 403), so all scores here are **CV-only**; Feb's best real submission (v3, on a non-canonical multi-seed scheme) posted Public LB AUC 0.95332 as a loose sanity anchor.

Models & Architecture

Purpose of Architecture. A binary heart-disease classifier that maximizes ROC-AUC. **Architecture Type** is a **gradient-boosted decision-tree ensemble** — LightGBM as the primary learner, with XGBoost and CatBoost pool members, combined by a weighted blend; the champion is a **37-member mega-blend** discovered at tree-search node #43.

Each member takes the same **Input Format** — a numeric feature matrix (no categorical encoding needed; all columns are already numeric) — of **Input Dimension 13 features** for the champion LGB family (33 for the XGB/CAT members); a 1-D vector with no spatial/sequence structure.

Architecture Description. Each GBDT member is an additive ensemble of decision trees grown by histogram-based, leaf-wise gradient boosting (representative stage-2 LGB: ~646 boosting iterations; the Optuna-tuned champion LGB uses `num_leaves` = 102 with a shallow `max_depth` = 4 and strong regularization). The blend layer sits on top as a convex weighted average of members' out-of-fold predictions, with weights found by Dirichlet sampling plus coordinate-ascent refinement. **Model Complexity** is therefore best expressed as trees × leaves rather than a dense parameter count: the champion blends 37 pool members, though the weight search kept only **8 non-zero members** — **all LGB-family variants** of the tuned base-13 LGB, so in practice the mega-blend is an intra-LGB-family variance reduction.

Training procedures

Training proceeds as a **four-stage ladder** on the shared 5-fold split, each stage measured on the same OOF scale:

Stage	Configuration	OOF AUC
1	Feb v1 LGB+XGB anchor (34 features, fold-identical, reused not retrained)	0.95498
2	skill pipeline: 33-feature LGB+XGB+CAT blend (weights 0.15 / 0.25 / 0.60)	0.955197
3	linear iteration: base-13 prior check + <code>is_unbalance</code> ablation + Optuna 40-trial tune + seed-bag, 5-way blend	0.955510
4	tree-search <code>harness_v3</code> (60 nodes, node #43 = 37-member mega-blend)	0.955529

Two experience-library **priors were re-verified by controlled same-folds comparison** in stage 3: (A) "explicit product/threshold columns add nothing for GBDT" (s3e9) was **confirmed** — the 13 raw features (0.955238) beat the full 33-feature LGB (0.955009) by +0.000229, the *opposite* verdict to s6e1 but the *same* as s5e10; (B) "class-imbalance weighting hurts a ranking metric" (s3e3/s4e1) was **confirmed a third time** — `is_unbalance` = True (0.954993) lost to the default (0.955009) by -0.000016. The **Loss Function** is binary log-loss (LightGBM `binary` objective) with model selection on AUC; the default **unweighted** objective was kept. The **Optimization Algorithm** is gradient boosting (histogram, leaf-wise) — **not** SGD/ADAM — with blend weights optimized by Dirichlet search + coordinate-ascent. The **Learning Rate** is the boosting shrinkage (tuned champion LGB `learning_rate` ≈ 0.0171); there is **no Learning Rate Scheduler** (N/A — *GBDT convergence is governed by CV early-stopping, not a schedule*) and **no Batch Size** (N/A — *full-dataset histogram boosting, not mini-batched*).

Training Duration for the stage-4 tree search was **~7,901 s of wall-clock over 60 evaluated nodes** (0 failed, 0 dedup rejections; the search stopped on the 60-node hard budget cap); the stage-3 Optuna tune was 40 trials / 515.1 s on a fold-0 proxy. **Training Memory** was not explicitly capped — a $630k \times 33$ float matrix is comfortable on the single arm64 CPU machine; peak was not separately recorded. **Transfer Learning** is N/A (*no pretrained weights*); its analogue here is **cross-season experience injection** — distilled priors from s3e3/s3e9/s5e10/s6e1 carried in and each re-verified on this data. **Data Augmentation** is N/A (*tabular*); the analogues are feature engineering and seed-bagging.

Reproducibility Standards are strict: fixed seeds (fold seed = 42; Optuna seed and seed-bag seed = 2024 logged), LightGBM determinism flags (`deterministic=True`, `force_row_wise=True`, `fixed_num_threads=16` across all models), pinned from the very first run (the hard-won s6e1 fix). This was a **native v3 run — no replay needed**; the stage-4 OOF-reproduction gate (`06_rebuild_tree_best.py`) retrained all 37 members and matched cached OOFs digit-for-digit ($\max|\Delta\text{OOF}| = 0$), recovering the blend AUC 0.955529 and the stored weights member-for-member.

Inference procedures

Decision Threshold is N/A — AUC is a ranking metric, so we submit raw positive-class probabilities and never threshold (a 0.5 cut would only be needed if hard labels were required). **Inference Duration** was not

separately profiled; GBDT scoring of the 270,000 test rows is sub-second per member on CPU, and **Inference Memory** is negligible (tree inference is lightweight, not capped).

Evaluation & Benchmarking

Task for Performance Evaluation. Rank the test patients by heart-disease probability. **Performance Metrics.** ROC-AUC is the sole competition metric; our stage-4 champion reaches **CV OOF AUC 0.955529** (submission file `0.95553`), climbing $0.95498 \rightarrow 0.955197 \rightarrow 0.955510 \rightarrow 0.955529$ across the four stages (stage $1 \rightarrow 4 \approx +0.0575\%$). The whole ladder moves within a ~ 0.0006 AUC band — an honest "last-mile" competition where gains are real but small, and the mega-blend still extracted a final $+0.000019$ over the linear stage's best blend. The Feb v3 prior-season submission (non-canonical multi-seed CV) posted Public LB 0.95332 — reference only, as this run is CV-only.

Performance Benchmarking. The natural comparator is the **NVIDIA reproduce-agent**, which copies the strongest public kernel:

Agent	Approach	CV AUC	Note
NVIDIA	reproduces <code>kospintr/heart-xgb-lightgbm-catb-baseline-k-fold</code> (verbatim-ish, CPU)	0.9554	tie
Our agent	from-scratch GBDT pool + tree-search v3	0.95553	nudges ahead

The result is effectively a **dead heat**, with our from-scratch agent fractionally ahead of the reproduced public kernel — consistent with the newer-season (S5–S6) pattern where the reproduce-vs-originate gap collapses, in contrast to the small public-kernel edge seen on older S3-era problems. Both numbers are leak-free.